

Solution

- (a) As the number of H-atoms undergoing ionization and recombination are balanced at R_S , each photon can ionize exactly one hydrogen atom and each neutral hydrogen has exactly one proton and one electron,

$$\begin{aligned}
 n_{\text{recomb}} &= n_{\text{HII}} = Q \\
 \text{and } n_e &= n_p = n_H \\
 n_{\text{recomb}} &= \alpha n_p n_e V_S \\
 Q &= \alpha n_H^2 \frac{4\pi}{3} R_S^3 \\
 \therefore R_S &= \sqrt[3]{\frac{3Q}{4\pi\alpha n_H^2}} \\
 &= \sqrt[3]{\frac{3 \times 10^{49}}{4\pi \times 10^{-19} \times (10^8)^2}} \\
 &= 1.3 \times 10^{17} \text{ m} \\
 \therefore R_S &\approx 4 \text{ pc}
 \end{aligned}$$

- (b) Per one unit length distance, a typical photon will encounter σn_H H-atoms. Thus, the mean free path will be,

$$\begin{aligned}
 l_{\nu_0} &= \frac{1}{\sigma n_H} = \frac{1}{\sigma 10^{-21} \times 10^8} \\
 l_{\nu_0} &= 10^{13} \text{ m} \\
 \therefore l_{\nu_0} &\approx 10^{-4} R_S \ll R_S
 \end{aligned}$$

Thus, the boundary layer of the sphere is very thin as compared to its total size. Hence,

“YES” this ionized nebula is very sharp-edged.

- (c) For Stromgren sphere to form, all H-atoms (N) inside R_S need to be ionized. Thus, time t_S required will be

$$\begin{aligned}
 N &= V_S n_H = \frac{4\pi}{3} R_S^3 n_H \\
 t_S &= \frac{N}{Q} = \frac{4\pi R_S^3 n_H}{3Q} \\
 &= \frac{1}{\alpha n_H} = \frac{1}{10^{-19} \times 10^8} \\
 &= 10^{11} \text{ s} \\
 &\approx 3000 \text{ yr}
 \end{aligned}$$

For typical nebular densities, the main-sequence lifetime of LBV stars is much longer than this ionization time, and hence our assumption of a stationary system is justified.